

Outstanding Improvement Quality Of Life In Severe Lumbar Central Canal Stenosis With Cauda Equina Syndrome Using Double Entry Percutaneous Laser Disc Decompression Combined With Umbilical Cord-Derived Mesenchymal Stem Cells Implantation: A Case Report

Ahmad Jabir Rahyussalim
Bima Adi Wiryo
Eko Perdana Putra
Tri Kurniawati
Bagus Wijaya Kusuma

Department of Orthopaedic and Traumatology, Cipto Mangunkusumo National General Hospital and Faculty of Medicine Universitas Indonesia, Jakarta, Indonesia

Stem Cell Medical Technology Integrated Service Unit, Cipto Mangunkusumo National General Hospital, Jakarta, Indonesia

Stem Cells and Tissue Engineering Research Cluster, Indonesian Medical Education and Research Institute (IMERI), Faculty of Medicine Universitas Indonesia, Jakarta, Indonesia

Introduction

Lumbar canal stenosis is a debilitating spinal condition causing significant pain, neurogenic claudication, and overall diminished quality of life

In severe cases, traditional decompression surgeries, such as open laminotomy, discectomy, foraminotomy Percutaneous Endoscopic Foraminoplasty can be invasive and have associated risks offer relief but can be invasive and carry associated risks

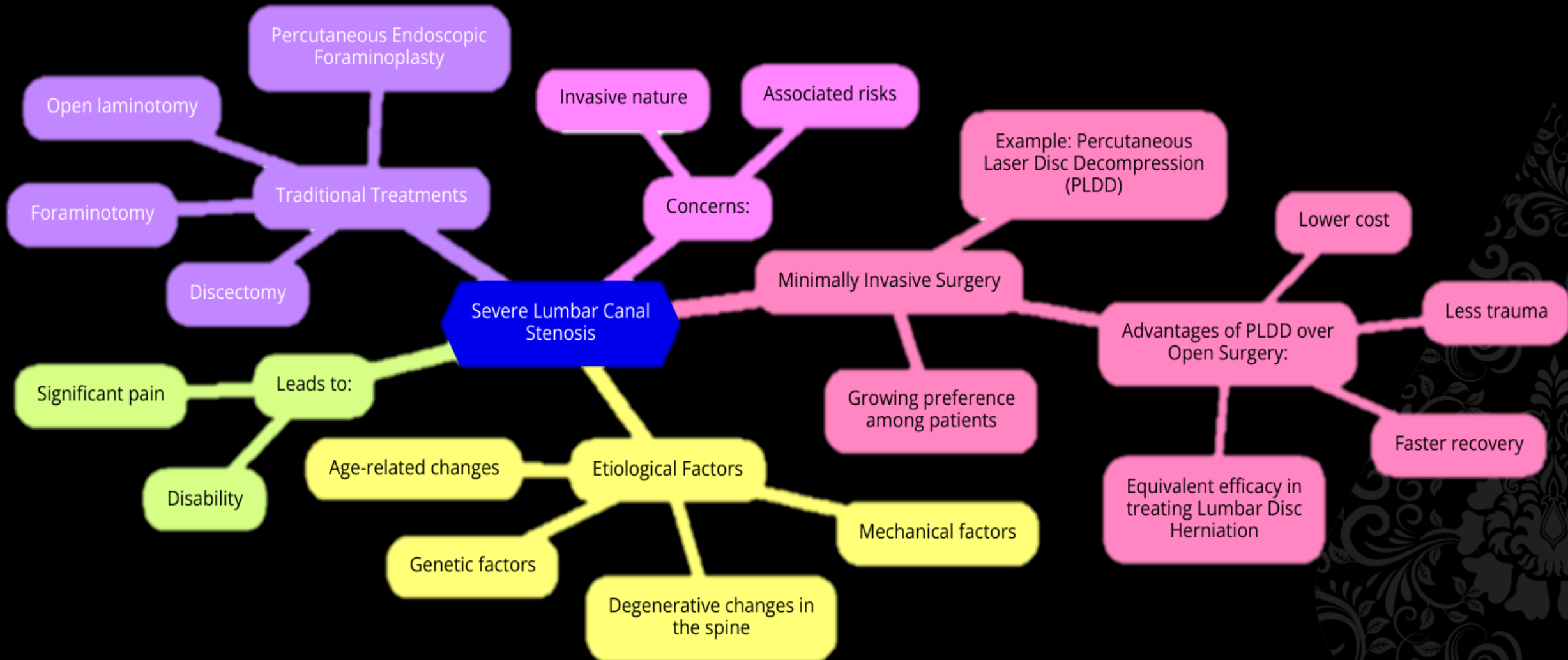
The epidemiology of LCS is characterized by a high prevalence in older adults, with men being more affected than women

In the US, LCS also plays a major role in spinal surgery; 5.9 out of 100 individuals have a lumbar fusion within a year of receiving a lumbar degeneration diagnosis.

Severe lumbar canal stenosis is a complex condition with multiple etiological factors



Introduction – Why?



Case Presentation



Figure 1. Pre-operation Pain Projection Chart showing Motoric, Sensoric, Bowel, Bladder, Dysfunction that affect quality of life. The area of pain in the back and buttocks is marked by a blue cross marker. There was also decreased motor function on a scale of 3 through bilateral legs.

- **A 31-year-old** male patient presented to the orthopaedic clinic with worsening back pain VAS 7 and weakness on both legs, resulting in chronic low back pain and progressive neurological deficits.
- He came with a 1-month history of progressive weakness in both lower extremities and worsening lower back pain, in the past 10 days - motor strength scale of 3
- He experienced lower extremity motor impairment, including numbness, stiffness, radiating pain, Bladder and Bowel problem resulting difficulty with daily activities due to decreased spinal mobility and radiating pain in the buttocks and hips.
- Unable to sit, Walk, Standing by himself which disrupts his quality of life.
- Unable to control urination and defecation, so he came wearing a diaper and a catheter

Pre - Implantation



Unable to sit

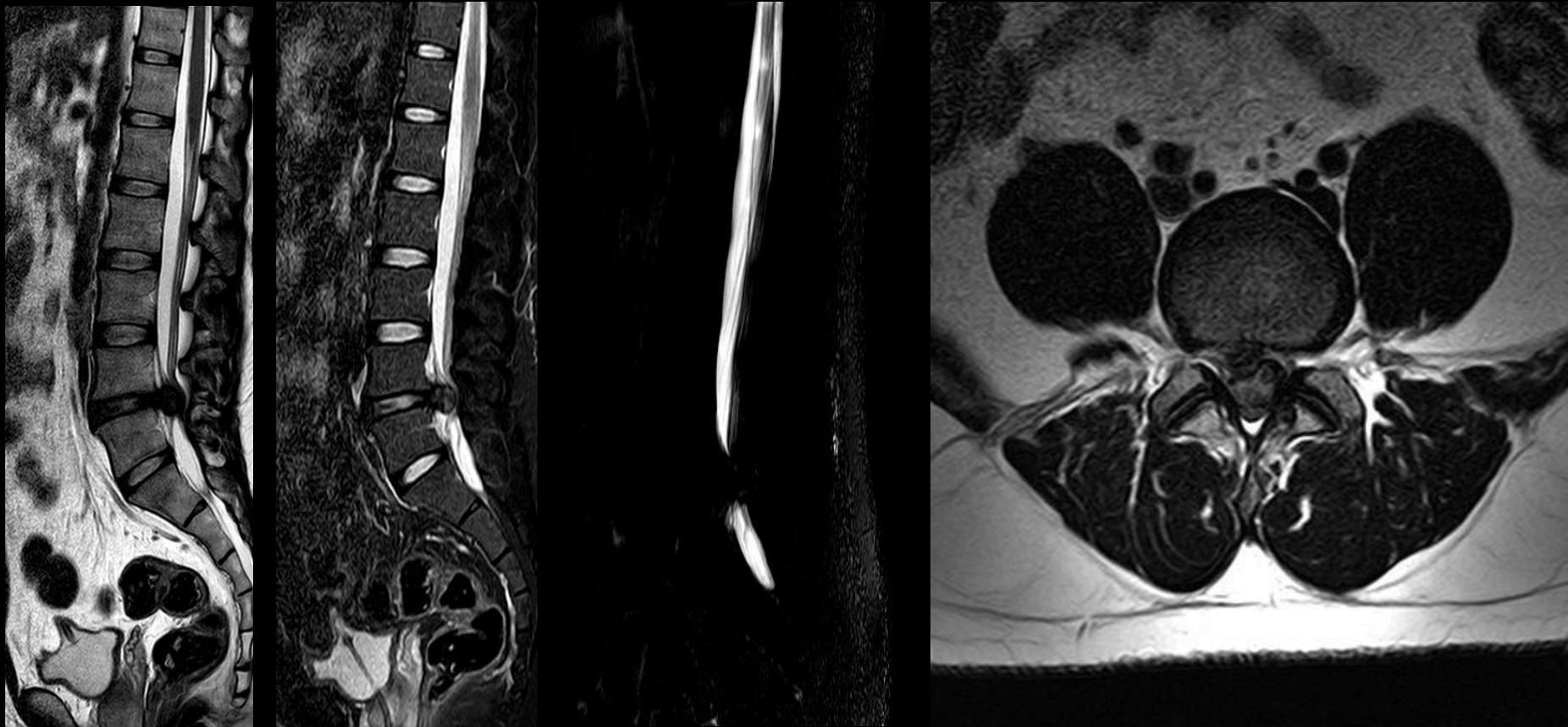


Unable to stand



Unable to walk

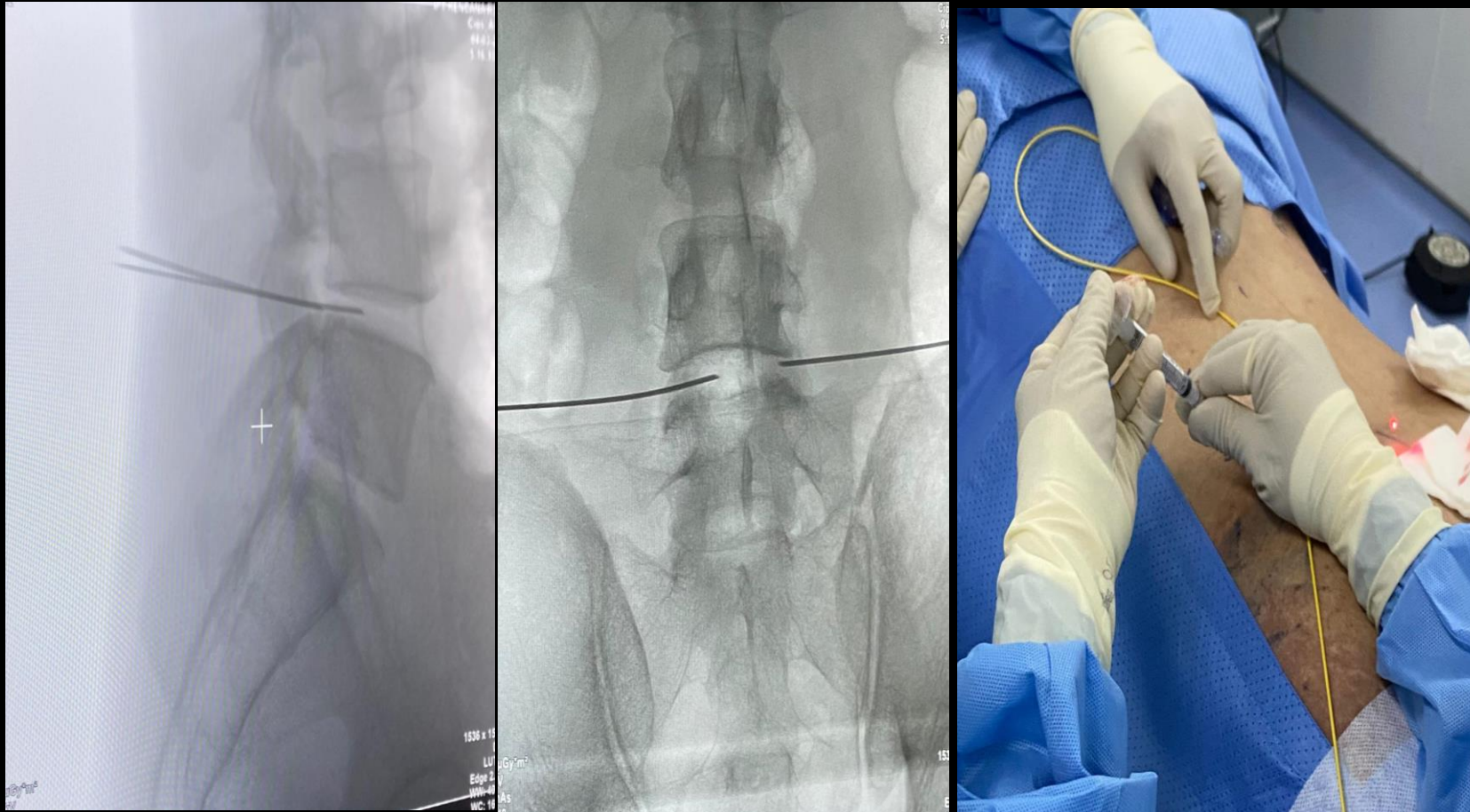
Case Presentation



- Magnetic resonance imaging (MRI) confirmed L5-S1 severe lumbar central canal stenosis with Giant HNP causing total blocking of the spinal cord and with multilevel disc degeneration.

Figure 1. Sagittal dan Axial T2 MRI Lumbosacral show L5-S1 severe total blocking canal stenosis followed by multiple disc degeneration pfirmann type II-III. The L4-L5 segment demonstrate disc herniation total compression on the thecal sac, and bilateral complete foraminal stenosis with pfirmann type V complete degeneration.

Intervention



- The patient underwent double entry percutaneous laser disc decompression direct into the lesion where the Severe Blocking L5-S1 Central Canal Stenosis, followed by local implantation of umbilical cord derived MSCs. (see Figure-3)
- Under the C-Arm Guidance the Local Implantation including direct implantation of MSCs (Stem cell and Secretome)
- Multiple Needling surgery performed without incision approach with no complication and fast recovery, Patient went back home the next day after the surgery

<24 hours after Implantation

Motoric
Significantly
Increase



Able to walk
No Catheter



20 days After Implantation



HNP LEVEL PARAH, L4-5
GA BISA PIPIS GA BISA EO



SENIN 26 FEB 2024



Scan to play videos
made by the patient
himself

20 days Post – Able to go shopping and going up
and down stairs

1,5 months After Implantation



- Able to walk faster
- Able to squat



3 Month After Implantation



Able to run, jump, and return to work and normal daily activities

Single Entry

Double Entry

Repeated needle adjustment and multi-point ignition -> increased intervertebral disc inflammation due to thermal injury

Excessive gas vaporization - increased pressure in the disc

More Area to Implanted the stem cells

Overflow gasification – avoid excessive pressure

Decompress the opposite side

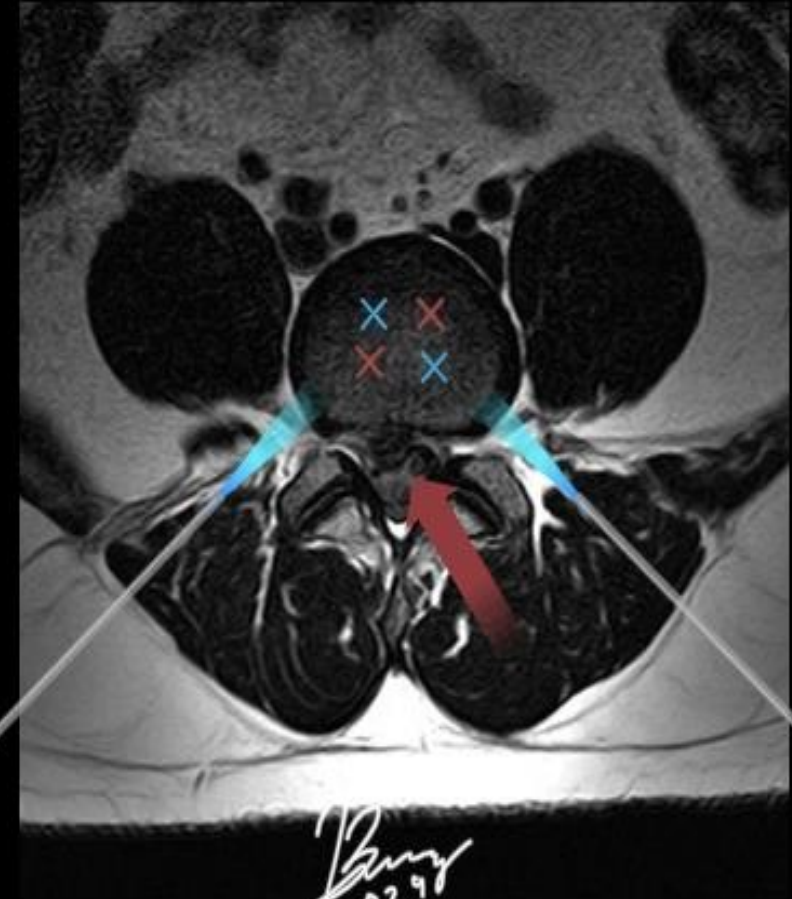


Illustration of Double Entry Percutaneous Laser Disc Decompression (PLDD) Combined With Umbilical Cord-Derived Mesenchymal Stem Cells Implantation under C-Arm Guidance. Note that the cross red and blue mark are the Intradiscal double entry Implantation

Results

<i>Sign & Symptoms</i>		<i>Results</i>	
		<i><24 Hours post operatively</i>	<i>3 months Follow Up</i>
Motoric	weakness in both legs, radicular pain in both lower extremities, Unable to sit, Walk, Standing, & Run by him self, ODI Score 94%	Able to slow walk and standing by holding	Able to drive a car, Return to Office for work, Able to sit, Run, Walk, Standing, jumping by him self, ODI score 16%
Sensorics	Burning sensation, numbness, tingling, thickness, in both lower extremities, Lower Back Pain VAS 6 usually when sitting	Burning gone, reduced Thickness, numbness, tingling in both extremities, Pain significantly reduce VAS 3	Thickness, numbness, tingling in both extremities have reduced drastically, pain disappear
Bladder	On Catheter, Unable to control urination	can urinate spontaneously	back to normal no complaints
Bowel	wear diapers, Unable to control bowel movements	Bowel control Improvement	back to normal no complaints
Sexual	Difficult to gain erection	n/a	Able to gain Erection

Why double Entry?

Author, Year	Title	Case summary (patients and diagnosis)	Intervention and outcome
Hashemi, M. et al. (2020)	The Long-term Effects of Percutaneous Laser Disc Decompression (PLDD) Treatment on Lumbar Disc Protrusion: A 2-Year Follow-up	40 patients experienced chronic low back pain diagnosed with disc protrusion by MRI with 2 years follow up	Two levels and one level of Lumbar PLDD was performed with significant improvement of pain and functional disability
Xie, P. et al. (2020)	Percutaneous transforaminal full endoscopic decompression for the treatment of Lumbar Spinal Stenosis	Retrospective study of 45 patients diagnosed with one level lumbar spinal stenosis	The patients underwent percutaneous transforaminal full endoscopic decompression resulting in improved pain and disability
Zeng, Z. et al. (2020)	Efficacy and safety of percutaneous laser disc decompression using double-needle realignment method for lumbar disc herniation.	A single centered retrospective analysis of 111 patients diagnosed with contained lumbar disc herniation	56 cases treated with double-needle realignment PLDD and 55 cases treated with single-needle PLDD. The efficacy, VAS back pain scores, and risk of surgery was better on double-needle realignment PLDD group.

Regenerative Properties – Philosophy

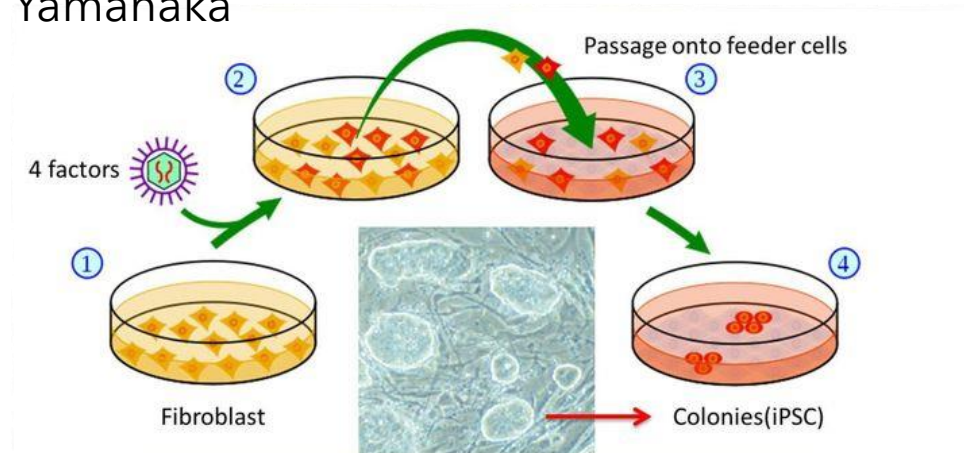
Reprogramming : Induced Pluripotent Stem Cells (iPSCs)



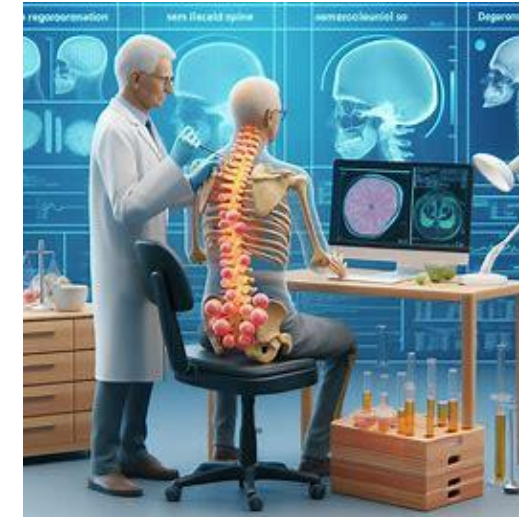
Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. [Cell](#). 2006 Aug 25;126(4):663-76. Epub 2006 Aug 10.

In 2012 **Yamanaka** and **John Gurdon** were awarded the [Nobel Prize for Physiology or Medicine](#) for the discovery that mature fibroblast can be converted to stem cells.

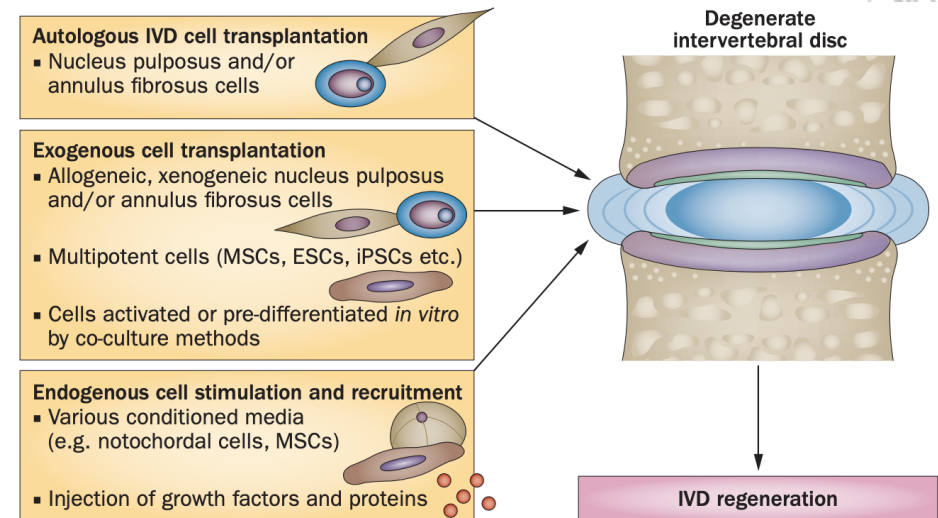
Shinya Yamanaka



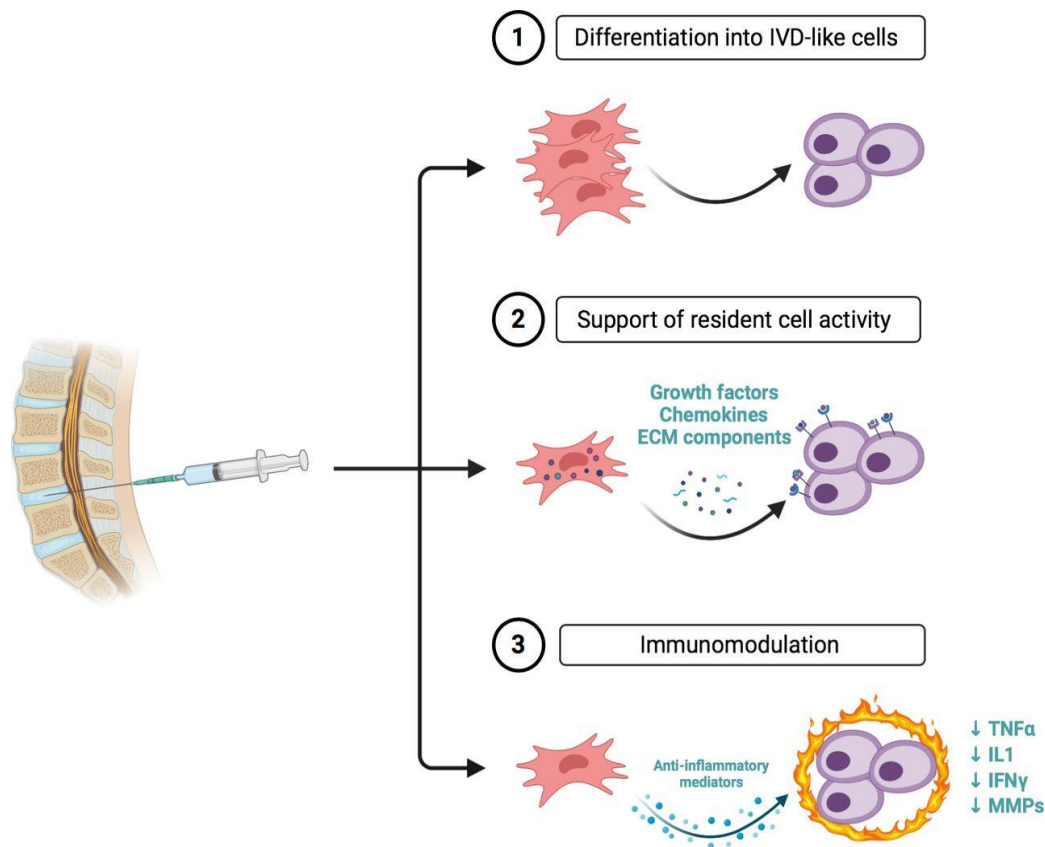
Future Of Medicine?



Degeneration > < Regeneration



Regenerative Properties – How do it works?



The main effects of stem cells following transplantation into the intervertebral disc. IVD, intervertebral disc; ECM, extracellular matrix; TNF α , tumor necrosis factor- α ; IL, interleukin; IFN, interferon; MMP, matrix metalloproteinase.

Authors	Type of Study	Year	No. of Patients	Inclusion Criteria	Cell Source	Cell Concentration	Outcomes	Follow-Up
Orozco et al ⁵⁹	Prospective	2011	10	18–65 yo with chronic LBP (≥ 6 mo) not responsive to conservative treatment with radiological signs of IDD and intact AF	Autologous BM-MSCs	$10 \pm 5 \times 10^6$ cells	- Improvement in VAS, ODI, and SF36 - Change in T2 MRI signal intensity within the NP	12 mo
Pettine et al ^{60,61}	Prospective	2015	26	Patients with chronic LBP (≥ 6 mo) not responsive to conservative treatment (Pfirrmann IV–VII)	Autologous BMAC		- Improvement in VAS and ODI - Change in T2 MRI signal intensity within the NP - Progression to surgery	24 mo
Elabd et al ⁶⁵	Prospective	2016	5	18–65 yo with chronic LBP (≥ 3 mo) not responsive to conservative treatment and a positive discography	Autologous BM-MSCs	$15.1\text{--}51.6 \times 10^6$ cells + autologous platelet lysate	- Improvement in clinical symptoms and quality of life - Change in MRI signal intensity within the NP	48–72 mo
Kumar et al ⁶²	Prospective	2017	10	19–70 yo with chronic LBP (≥ 3 mo) not responsive to conservative treatment (Pfirrmann III–IV)	Autologous ADSCs	25×10^6 cells + HA derivative 40×10^6 cells + HA derivative	- Rate of SAE - Improvement in VAS, ODI, and SF36 - Change in MRI signal intensity within the NP	12 mo
Comella et al ⁶³	Prospective	2017	15	18–90 yo with chronic LBP (≥ 6 mo) not responsive to conservative treatment and intact AF	SVF	$30\text{--}60 \times 10^6$ cells in 1–3 mL + autologous PRP	- Rate of SAE - Improvement in VAS, PPI, BDI, ODI, SF12, and Dallas pain questionnaire	6 mo
Noriega et al ⁶⁴	RCT	2017	24	18–75 yo with chronic LBP (≥ 6 mo) not responsive to conservative treatment with radiological signs of IDD and intact AF	Allogeneic BM-MSCs	25×10^6 cells in 2 mL vs Sham infiltration	- Improvement in VAS, ODI, and SF12 - Change in T2 MRI signal intensity within the NP	12 mo
Amirdelfan et al ⁶⁶	RCT	2020	100	> 18yo with chronic LBP (≥ 6 mo) not responsive to conservative treatment with moderate IDD (modified Pfirrmann III–VI)	Allogeneic STRO-3 ⁺ MPCs	8×10^6 cells + 1% HA vs 16×10^6 cells + 1% HA vs 1% HA only vs Sham infiltration	- Rate of SAE - Improvement in VAS, ODI, SF36 and WPAI - Change in T2 MRI signal intensity within the NP	36 mo

Abbreviations: ADSCs, adipose-derived stem cells; AF, annulus fibrosus; BDI, Beck Depression Inventory; BM-MSCs, bone marrow-derived mesenchymal stem cells; BMAC, bone marrow aspirate concentrate; HA, hyaluronic acid; IDD, intervertebral disc degeneration; LBP, low back pain; MPCs, mesenchymal precursor cells; MRI, magnetic resonance imaging; NP, nucleus pulposus; ODI, Oswestry Disability Index; PPI, present pain intensity; PRP, platelet-rich plasma; RCT, randomized controlled trial; SAE, serious adverse events; SF, Short Form Health Survey; STRO-3, stromal precursor antigen-3; SVF, stromal vascular fraction; T2, T2-weighted imaging; VAS, visual analog scale; WPAI, Work Productivity and Activity Index; yo, year-old.

Discussion



The procedure of percutaneous laser disc decompression has emerged as a minimally invasive technique for treating lumbar disc herniation, offering a **viable alternative** to traditional open surgery



PLDD involves the use of a laser fiber that is inserted into the intervertebral disc under fluoroscopic guidance, where the laser energy is applied to vaporize and shrink the herniated disc material



This targeted approach allows for decompression of the nerve root and reduction of pain, while minimizing damage to surrounding tissues

Discussion



Additionally, the minimally invasive nature of PLDD results in **minimal tissue injury, reduced operative time, shorter hospital stays, and quicker recovery times** compared to endoscopy or open surgery, making it a promising option for patients seeking relief from chronic back pain



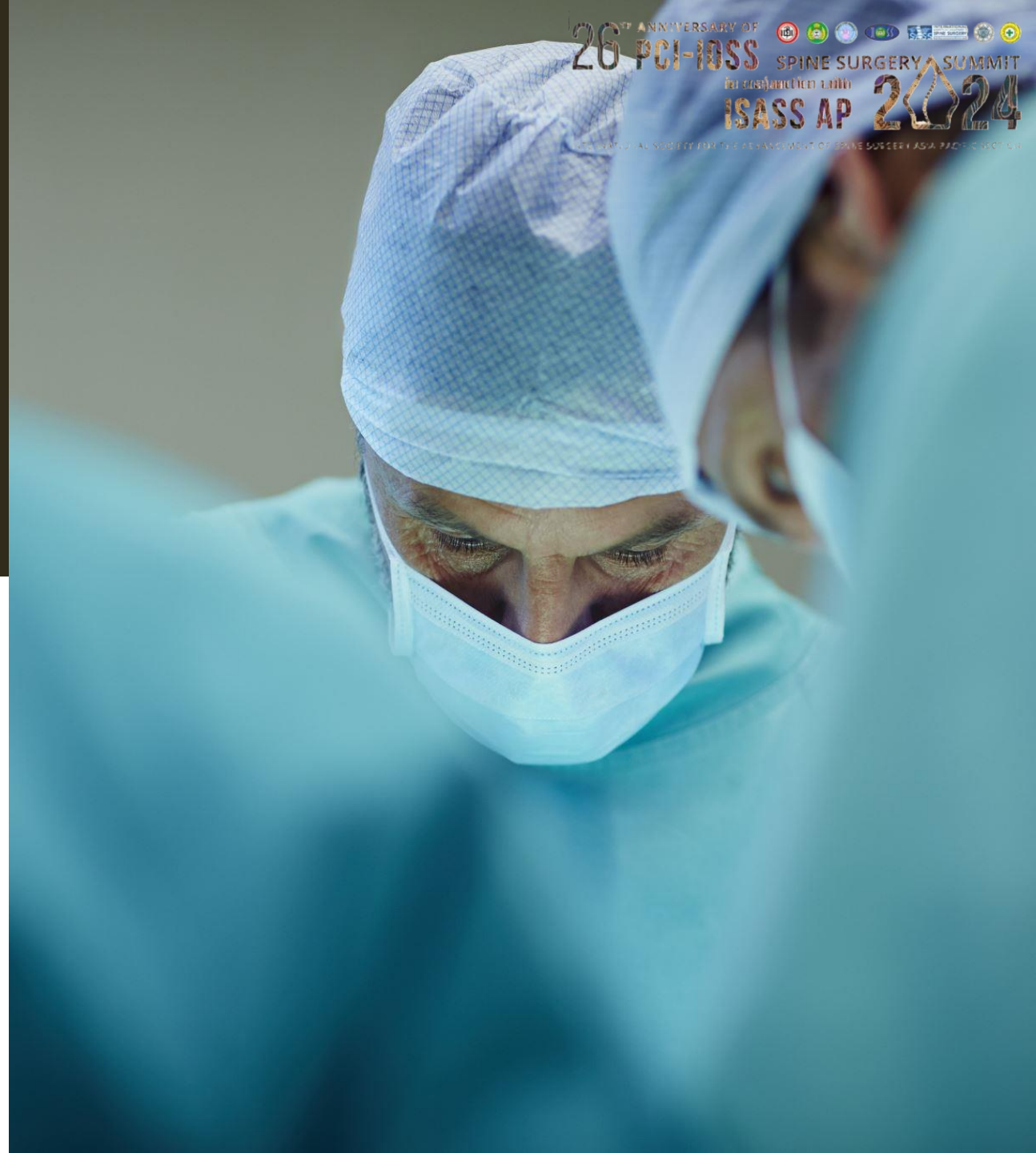
However, it is crucial to consider that while PLDD is a minimally invasive procedure with advantages such as **no skin incision and reduced anesthesia requirements**, it may have varying success rates and reoperation rates compared to traditional microdiscectomy.



the comparison of PLDD with conventional microdiscectomy for sciatica, highlighting the benefits of PLDD in terms of **lower healthcare costs and comparable or even better outcomes**

Discussion

Intradiscal injection of stem cells has the potential to **reverse or slow disc degeneration**, and therefore, may be an alternative to disc arthroplasty and spinal fusion procedures for the treatment of lower back pain



Results



This case study highlights the **potential benefits** of combining percutaneous laser disc decompression with stem cell implantation for severe lumbar canal stenosis



The procedure aims to alleviate pain, improve function, and ultimately **enhance the quality of life** for patients suffering from this condition



The case report analysis detailed the significant improvements in a patient's quality of life before and after undergoing percutaneous laser disc decompression and stem cell implantation for severe lumbar canal stenosis

Conclusion and Recommendation



This case report suggests that double entry PLDD combined with UC-MSD implantation is a **safe and effective approach** for managing severe lumbar canal stenosis, leading to significant improvement in quality of life



In addition to **advanced mechanical treatment, biological treatment with MSD** injections is believed to be a potent combination providing a new avenue for the management of severe LCCS



In conclusion, the combination of percutaneous laser disc decompression and stem cell implantation has shown promising results in improving the quality of life for patients suffering from severe lumbar canal stenosis

Conclussion and Recomendation

This innovative approach not only addresses the underlying **structural issues** of disc herniation but also **promotes tissue regeneration and healing** through the use of stem cells

The advantages of minimally invasive surgery over open surgery include reduced trauma, decreased bleeding, quicker recovery, shorter hospital stays, lower costs, improved spinal stability, and fewer problems



Clinical Importance



This case report highlights the importance of a **minimally invasive approach combining double entry PLDD** with UC-MSD implantation for treating severe LCCS. While traditional surgical decompression offers relief, it's often invasive and carries potential complications.



Additionally, the study found that patients who underwent percutaneous laser disc decompression and stem cell implantation for severe lumbar canal stenosis experienced significant improvements in pain relief, functional mobility, and overall quality of life.



Furthermore, the findings highlight the potential of combining advanced technology like laser disc decompression with regenerative medicine techniques such as stem cell therapy to address complex spinal conditions effectively.

Acknowledgements, References, and Contact Information:

The author thanks Dr. dr. Rahyussalim, SpOT(K) with case report conceptualization and executor of the case, Tri Kurniawati for supervision of the case, and the patient's willingness and compliance with participation in this case report.

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